>_ PEHT · PROFESSIONAL ETHICAL HACKING TRACK

MODULE 02

Building the Practice Lab

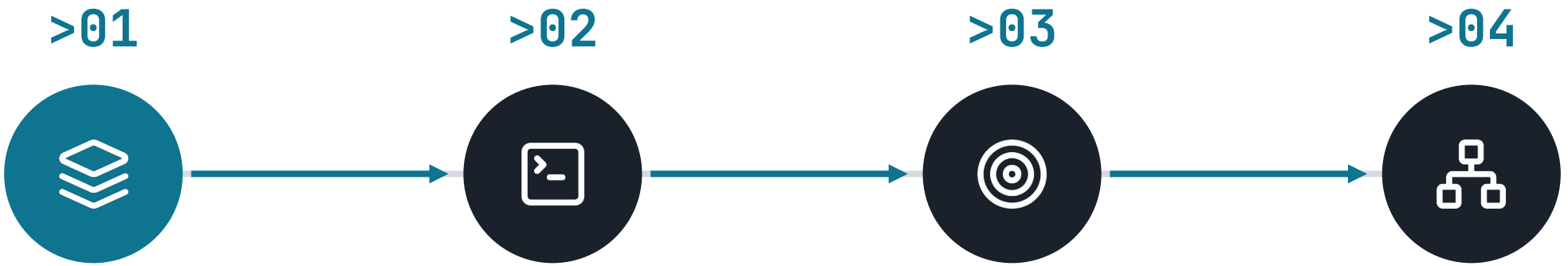
INSTRUCTOR

Mo'tasem Abu Baha'

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>_ MODULE 02 ROADMAP

Four lessons, one working lab



Preparing a Virtualization Environment

Installing a Penetration Testing OS

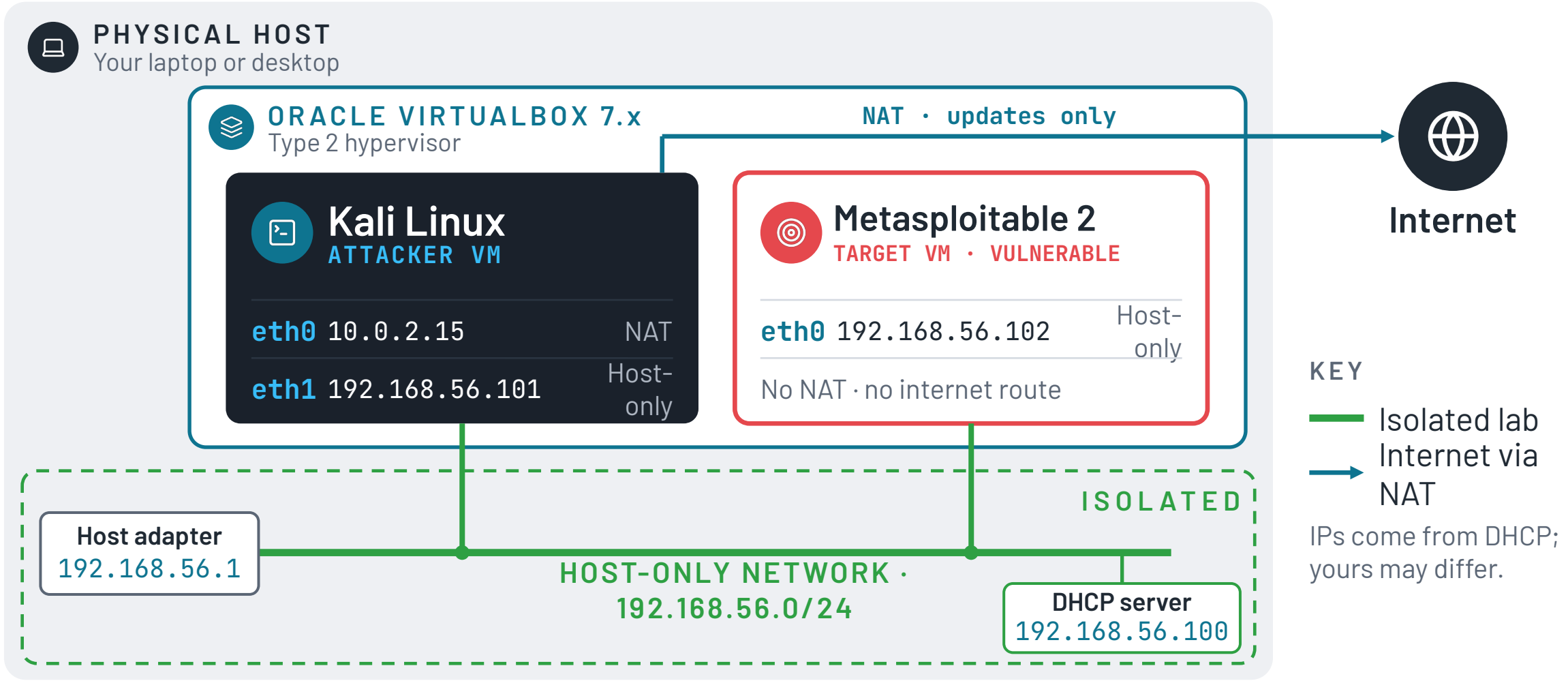
Deploying Vulnerable Target Machines

Configuring Isolated Network Environments

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>_ THE LAB YOU WILL BUILD

One host, two VMs, one isolated network



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>_ LEARNING OBJECTIVES

By the end of this module, you will...

+01 Prepare a host for virtualization

+02 Install and update Kali Linux

+03 Deploy Metasploitable 2 safely

+04 Isolate lab traffic with Host-only

+05 Verify, snapshot and restore your VMs

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LESSON 01

Preparing a Virtualization Environment

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Why build your own lab?



Legal

You own every target



Safe

Nothing leaves your host



Repeatable

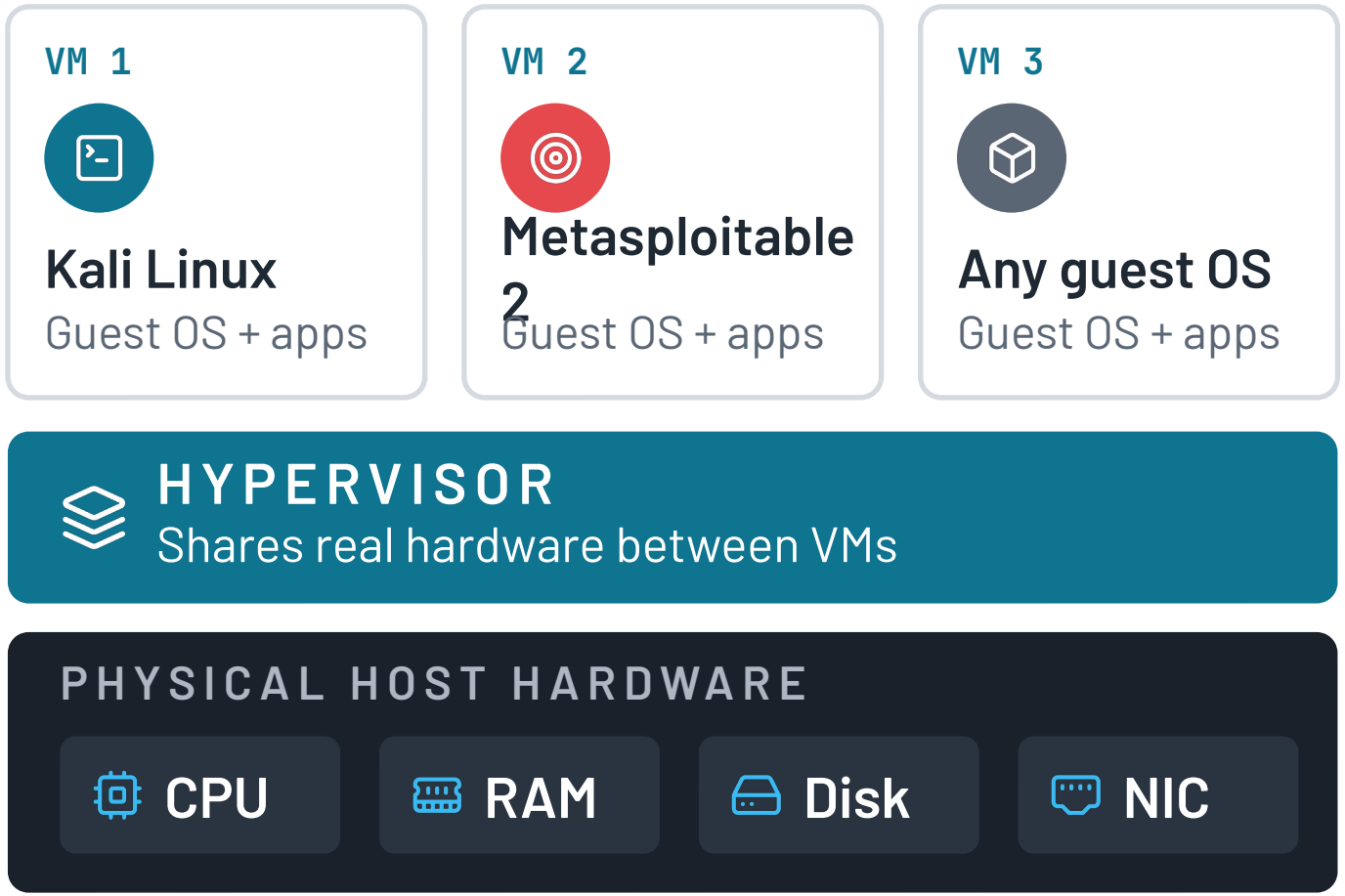
Reset and retry anytime



Free to break

Mistakes cost nothing

Virtualization: one computer, many machines



+ One PC runs many machines
+ Each VM: own OS, disk, NIC
+ VMs isolated from each other

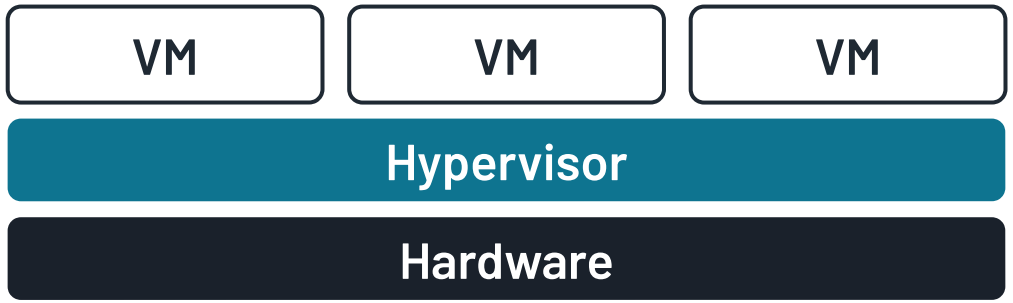
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Type 1 vs. Type 2: where does it run?

TYPE 1

Bare-metal

Runs on the hardware

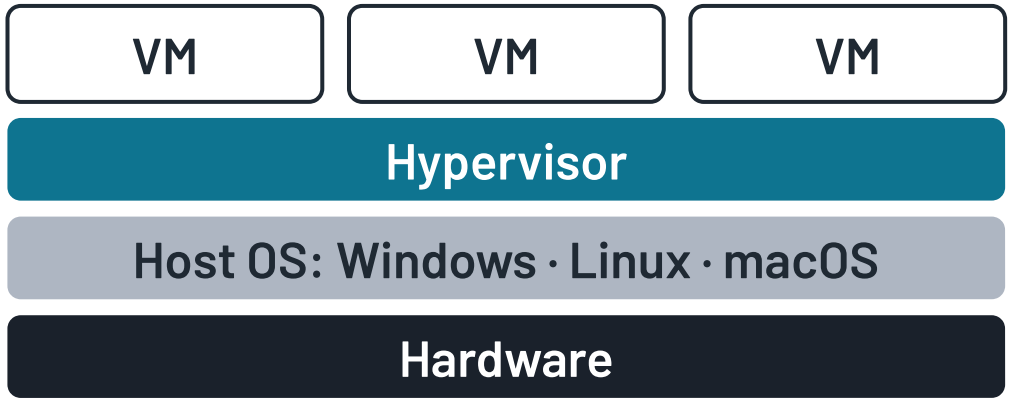


VMware ESXi · Proxmox VE · Hyper-V

TYPE 2 — OUR LAB

Hosted

Runs on a host OS



VirtualBox · VMware Workstation

Can your computer run the lab?

RESOURCE	MINIMUM	RECOMMENDED
CPU	64-bit, VT-x / AMD-V	**4+ cores**
RAM	8 GB	**16 GB**
Free disk	80 GB	**100 GB+**
Disk type	HDD works	**SSD**

RAM BUDGET · 16 GB HOST



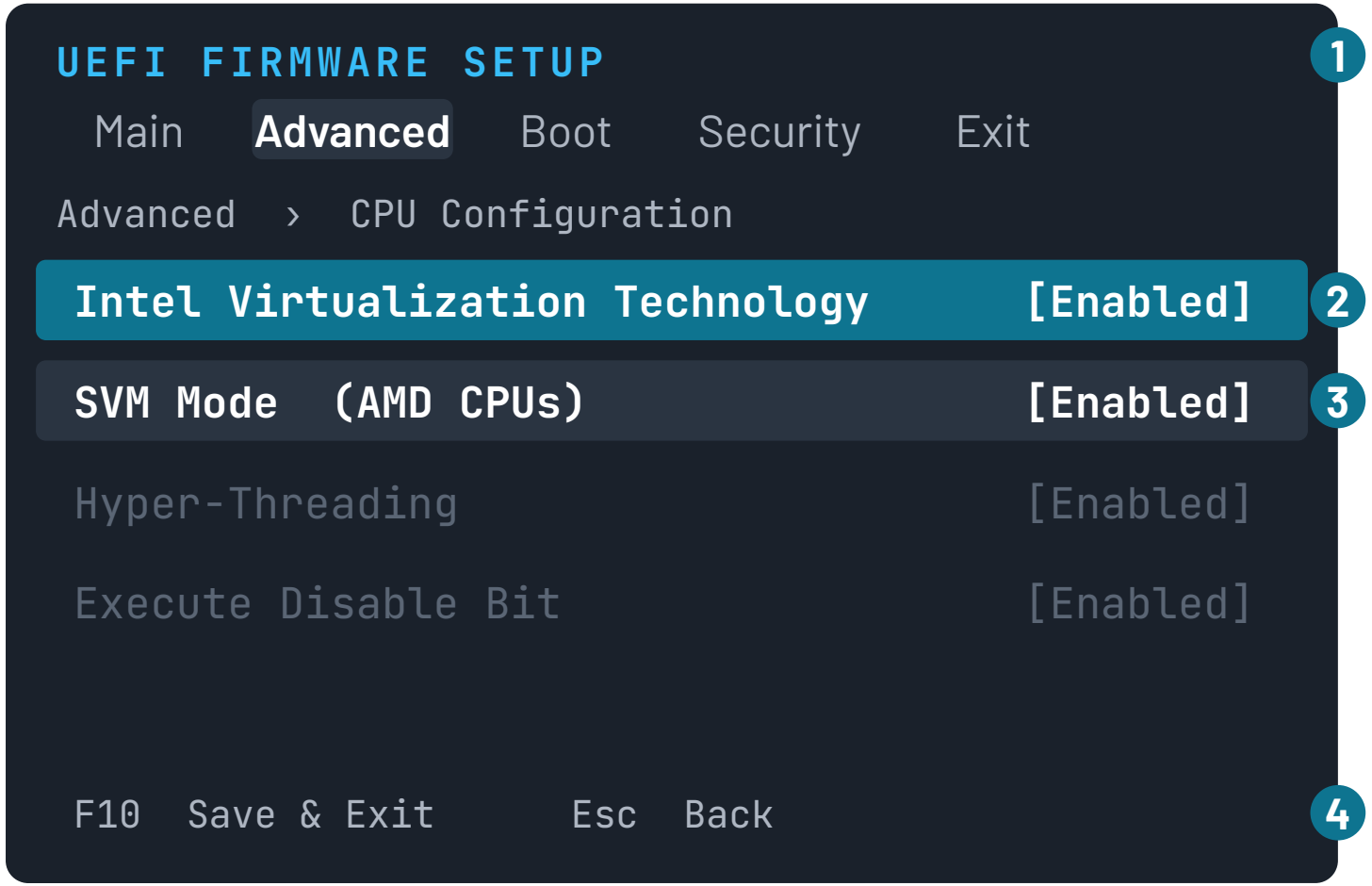
Headroom 3 GB

Metasploitable 1 GB

Kali Linux 4 GB

Host OS + apps 8 GB

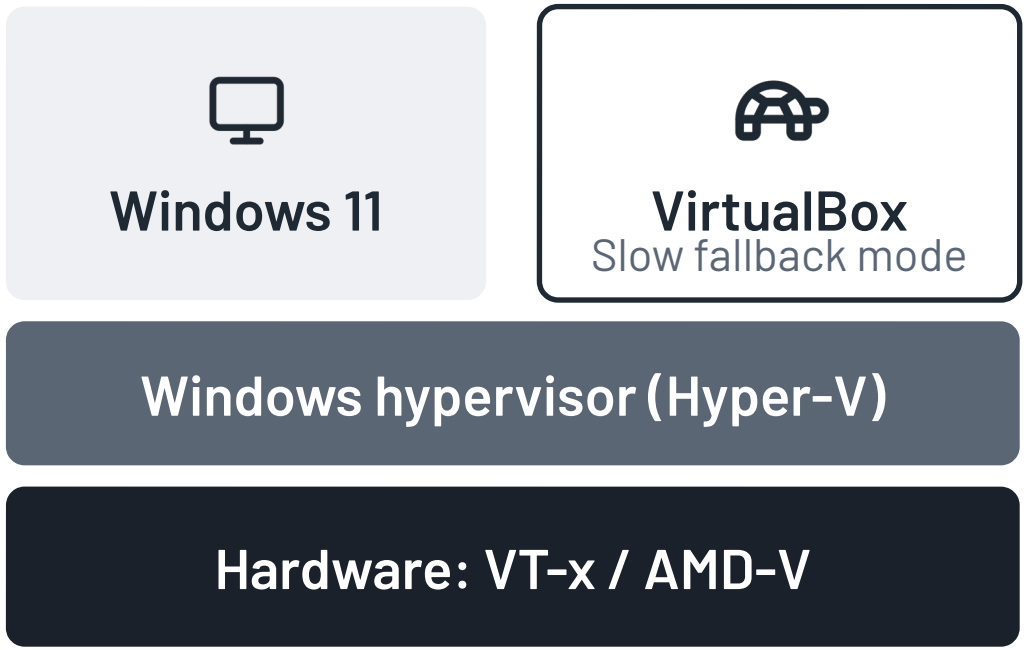
Switch on VT-x / AMD-V in firmware



1. **Enter setup at boot**
   F2, Del or Esc
2. **Intel CPU**
   Virtualization Technology → Enabled
3. **AMD CPU**
   SVM Mode → Enabled
4. **Save and reboot**
   F10

Windows conflict: Hyper-V holds the CPU

Hyper-V owns VT-x;
VirtualBox borrows it



Windows Features 1

Turn Windows features on or off

- ☐ Hyper-V
- ☐ Virtual Machine Platform
- ☐ Windows Hypervisor Platform

PowerShell · Administrator 2

```
PS> bcdedit /set hypervisorlaunchtype off
PS> shutdown /r /t 0
```


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Install VirtualBox 7.x, then its Extension Pack

1 

Download VirtualBox 7.x

virtualbox.org

2 

Run the installer

Accept driver prompts

3 

Get the Extension Pack

Exact same version

4 

Install the pack

Double-click .vbox-extpack

`VirtualBox 7.x.y` = `Extension Pack 7.x.y` 

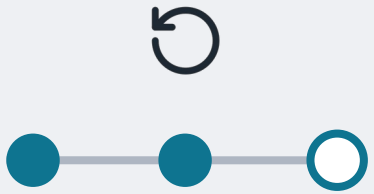
Versions must match

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Four VirtualBox features you'll use daily

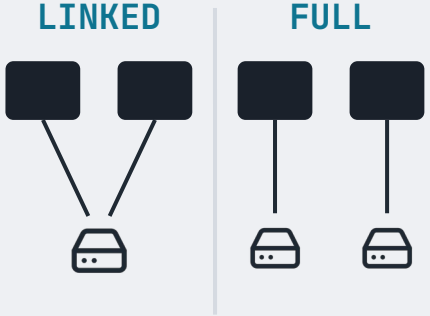
Snapshots

Save state, roll back



Clones

Linked or full copy



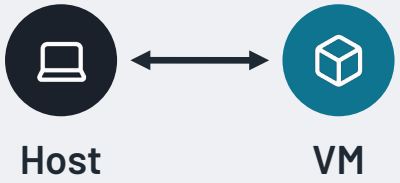
Guest Additions

Better display, drivers



Shared clipboard

Copy-paste host → VM



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Build your virtualization base



1. Confirm VT-x / AMD-V is enabled ☐
2. Windows: turn off Hyper-V features ☐
3. Install VirtualBox 7.x ☐
4. Install the matching Extension Pack ☐
5. Check Help › About VirtualBox ☐

>01 KEY TAKEAWAYS

What to remember from Lesson 01



+01

Hardware virtualization must be switched on



+02

VirtualBox is a Type 2 hypervisor



+03

Extension Pack version = VirtualBox version

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LESSON 02

Installing a Penetration Testing Operating System

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Why Kali Linux?



Debian

Base: familiar apt packages



600+

Security tools, ready to use



OffSec

Maintains and funds Kali

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Two ways in: pre-built image or ISO

PATH A

RECOMMENDED

Pre-built image

- Minutes to a desktop
- Preset, tuned defaults
- `kali / kali`

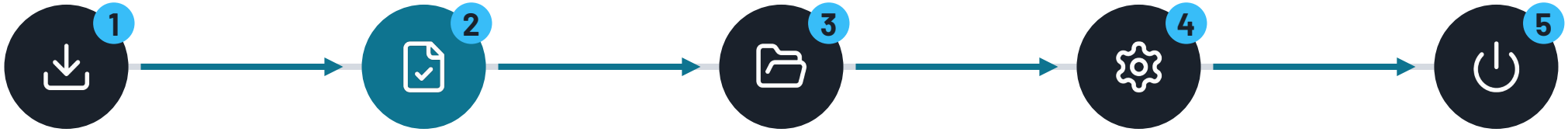
PATH B

ISO installer

- Full guided install
- You choose everything
- Set during install

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From download to first boot



Download
VirtualBox .7z from kali.org

Verify SHA256
Must match kali.org

Add to VirtualBox
Extract, then Machine › Add…

Configure
CPU, RAM, network

Boot
Log in: kali / kali

VERIFY THE DOWNLOAD

Linux host

```
$ sha256sum kali*.7z
```

Windows host · PowerShell

```
PS> Get-FileHash .\kali*.7z
```


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Recommended Kali VM settings

Kali Linux · Details

General

Name:	Kali Linux
Operating System:	Debian (64-bit)

System

Base Memory:	**4096 MB**	1
Processors:	**2**	2

Display

Video Memory:	**128 MB**	3
Graphics Controller:	**VMSVGA**	

Storage

SATA Port 0:	**kali-linux-...-amd64.vdi (80 GB)**	4

1. **RAM: 4 GB**
   2 GB minimum
2. **CPU: 2+ vCPUs**
   More helps tool speed
3. **Video: 128 MB**
   Graphics: VMSVGA
4. **Disk: 80 GB**
   Dynamically allocated

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First boot: new password, full update

kali@kali: ~

```
$ passwd
$ sudo apt update && sudo apt full-upgrade -y
$ sudo reboot
```

1

2

3

1

New password

Default kali / kali is public

2

Full upgrade

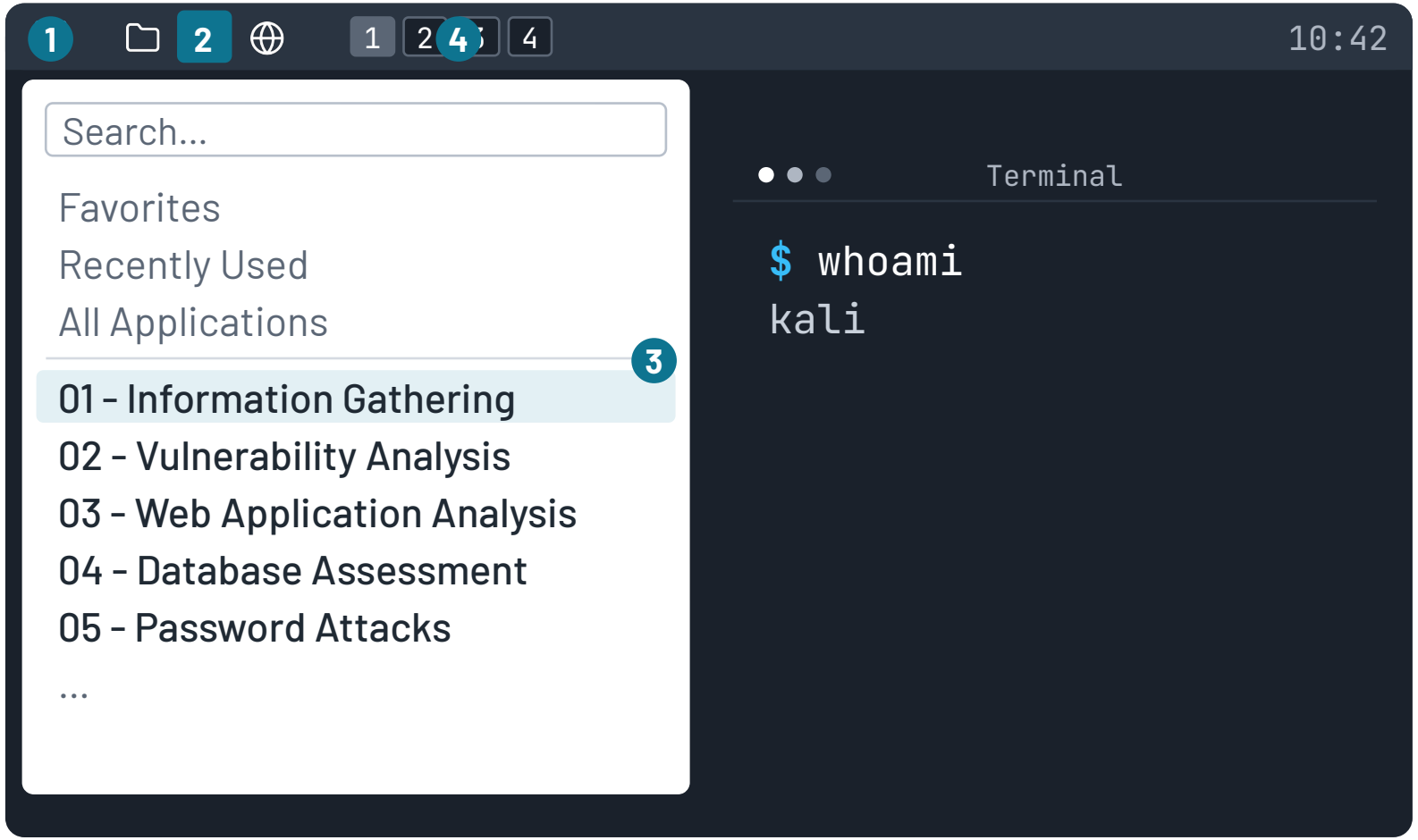
Refresh lists, then upgrade

3

Reboot

Load the new kernel

Find your way around the Kali desktop



1 Apps menu
Every tool, searchable

2 Terminal
Your main workspace

3 Tool categories
Grouped by attack phase

4 Workspaces
Four virtual desktops

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Snapshot the clean install

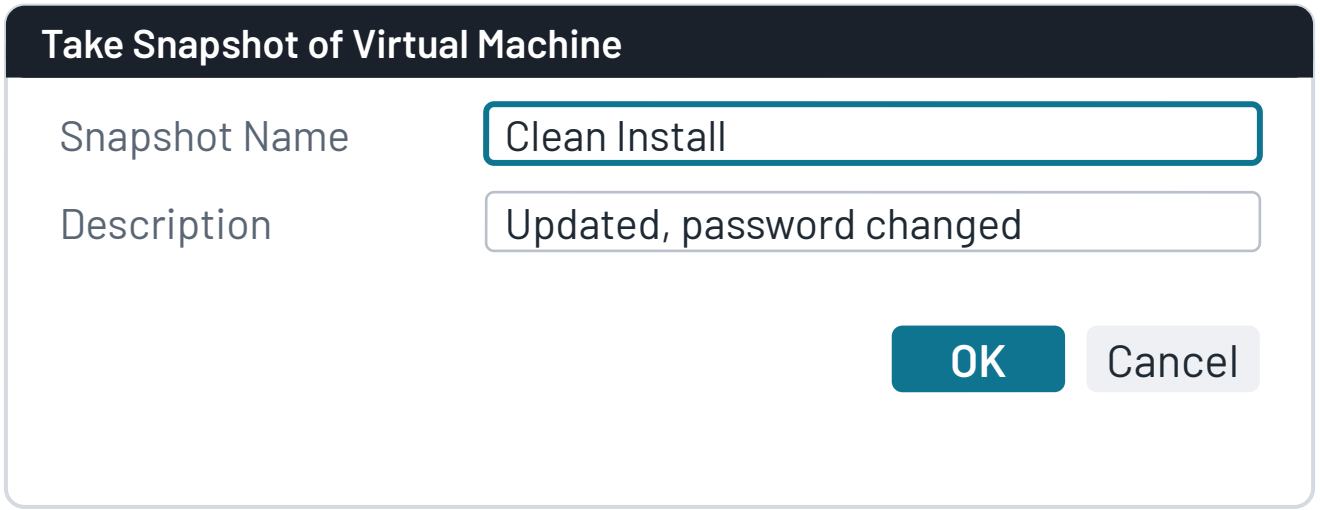


Imported
Fresh image

Clean Install
Updated, password set

Lab Ready
Lesson 04

Current State
Live VM



Machine › Take Snapshot…

Host + T

Host key: Right Ctrl (Windows, Linux)

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Get Kali running and clean



1. Download and verify the Kali image ☐
2. Add it to VirtualBox ☐
3. Apply the recommended CPU, RAM, video ☐
4. Change password, run full upgrade ☐
5. Take a “Clean Install” snapshot ☐

>02 KEY TAKEAWAYS

What to remember from Lesson 02



+01

Pre-built image: fastest, cleanest start



+02

Replace kali / kali on first boot



+03

Snapshot before you break things

LESSON 03

Deploying Vulnerable Target Machines

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Why attack a machine built to be broken?

Systems you don't own

Illegal without written permission

- Employer
- Websites
- Wi-Fi

VS

Your vulnerable VM

Legal, isolated, resettable

- Known flaws
- Host-only
- Snapshots

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Meet Metasploitable 2



Metasploitable 2

VULNERABLE BY DESIGN

PUBLISHER
Rapid7

BASE OS
Ubuntu 8.04 LTS

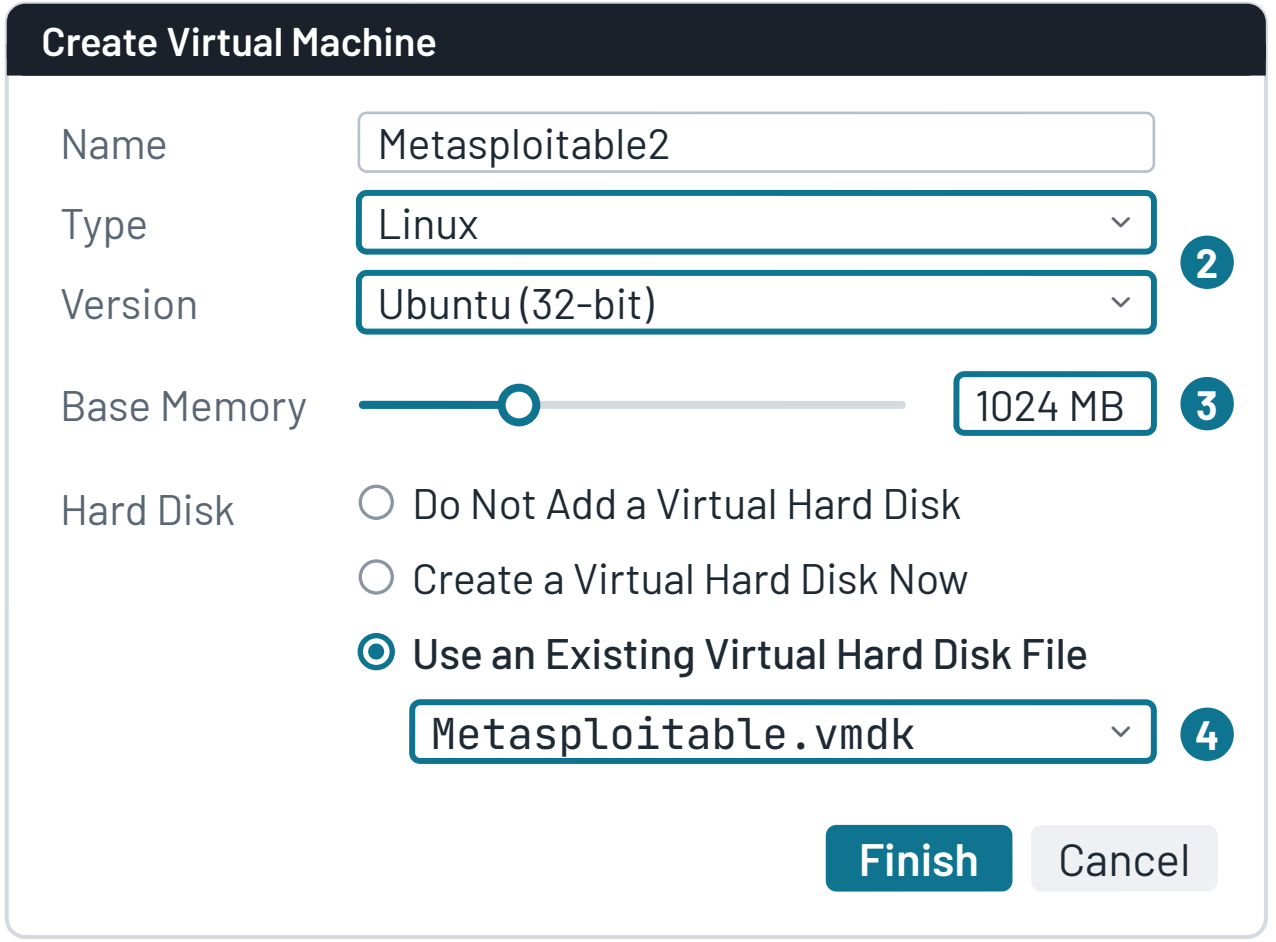
ARCHITECTURE
32-bit (x86)

DISK FORMAT
VMDK (VMware disk)

DEFAULT LOGIN
msfadmin / msfadmin

Deploy it in VirtualBox

1. Download and extract the zip
2. New VM: Linux, Ubuntu (32-bit)
3. Memory: 512 MB – 1 GB
4. Use existing disk: Metasploitable.vmdk
5. Adapter 1 → Host-only
6. Boot it



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What is listening on the target?

PORT	SERVICE	VERSION
21	FTP	• **vsftpd 2.3.4**
22	SSH	OpenSSH 4.7p1
23	Telnet	Linux telnetd
80	HTTP	Apache 2.2.8

PORT	SERVICE	VERSION
139/445	SMB	Samba smbd 3.0.20
3306	MySQL	MySQL 5.0.51a
5432	Postgres	PostgreSQL 8.3
6667	IRC	• **UnrealIRCd 3.2.8.1**

• Known backdoored release

Module 03 preview

```
$ sudo nmap -sV 192.168.56.102
```


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Five web apps waiting on port 80



DVWA

Damn Vulnerable Web App

`admin / password`



Mutillidae

OWASP practice targets

`/mutillidae`



phpMyAdmin

MySQL web console

`/phpMyAdmin`



TWiki

Legacy wiki platform

`/twiki`



WebDAV

File upload directory

`/dav`

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First login: find the target's IP

Metasploitable 2 · console

```
metasploitable login: msfadmin
Password:
$ ifconfig eth0
          inet addr:192.168.56.102  Bcast:192.168.56.255
```

ifconfig, not ip

Ubuntu 8.04 has no iproute2 ip tool

Note this IP

Your target address for every scan

Never put this VM on the internet



DANGER

Keep Metasploitable on Host-only

Never Bridged. Never NAT. Never a real network.



Bridged

On your real LAN

Exposed ✕



NAT

Routes to the internet

Exposed ✕



Host-only

Reachable only from Kali

Isolated ✓

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Stand up the target



1. Download and extract Metasploitable 2 ☐
2. Create the VM with the existing VMDK ☐
3. Set Adapter 1 to Host-only ☐
4. Boot and log in: msfadmin / msfadmin ☐
5. Run ifconfig and note the IP ☐

What to remember from Lesson 03



+01

Vulnerable VMs are legal, resettable targets



+02

Attach the VMDK; don’t create a new disk



+03

Host-only only — never Bridged or NAT

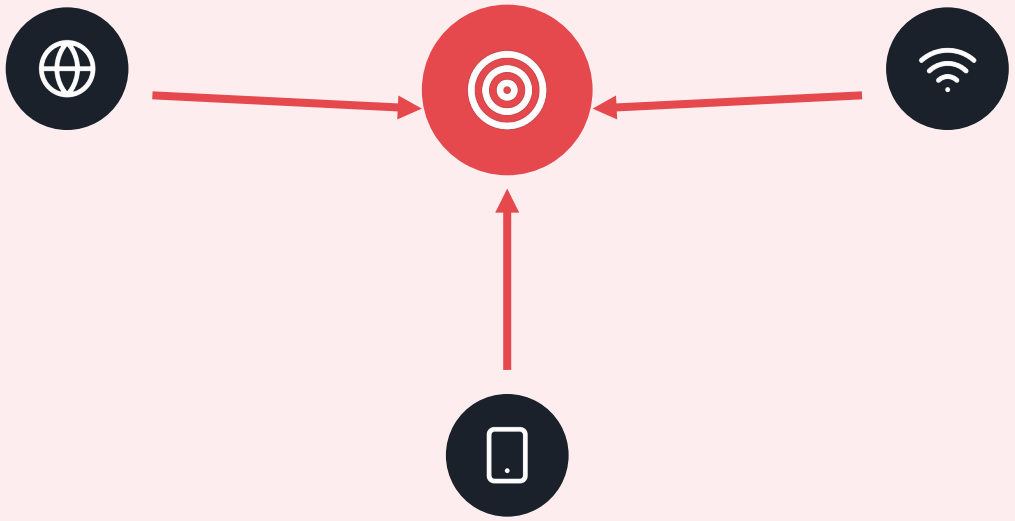
LESSON 04

Configuring Isolated Network Environments

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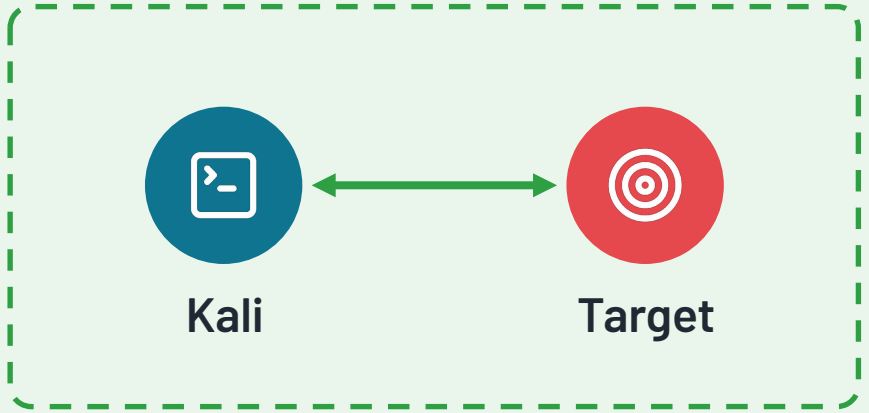
Why isolation decides everything

EXPOSED
Bridged adapter



Any device can reach the target

ISOLATED
Host-only adapter



Only your Kali VM can reach it

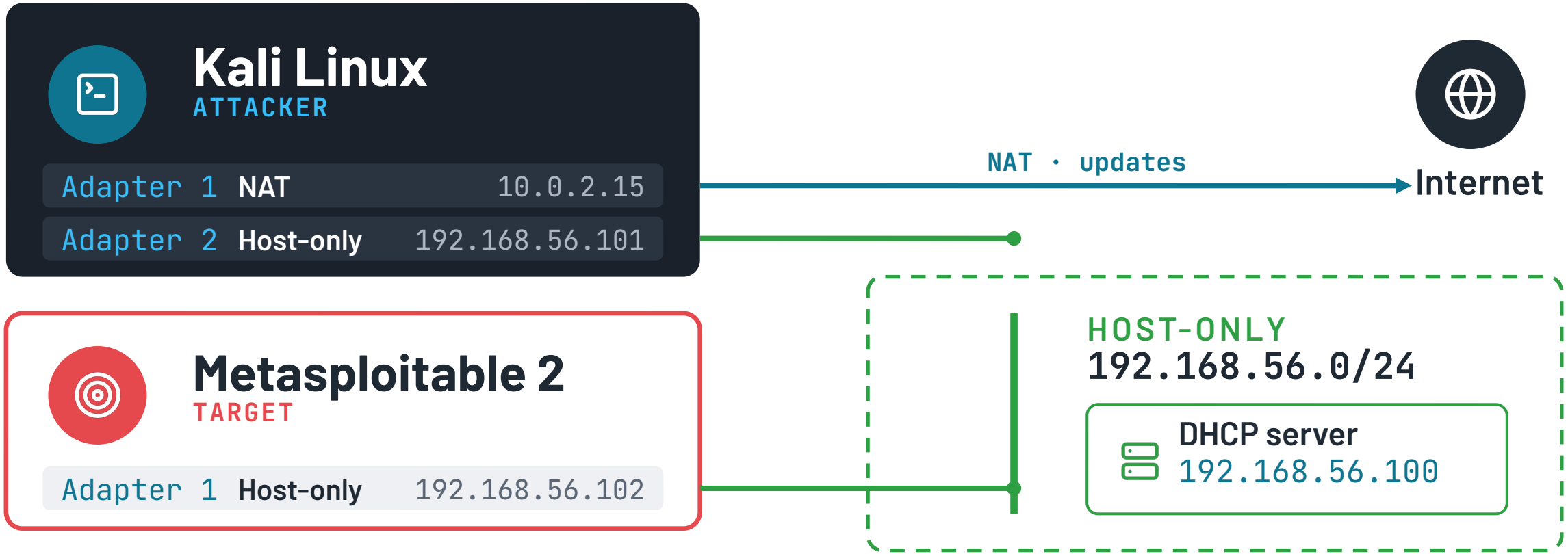
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Five network modes, one safe choice

MODE	VM → VM	VM → HOST	VM → NET	LAB SAFETY
NAT	✗	✗	✓	Safe · no inbound
NAT Network	–	–	✓	Safe · shared NAT
Bridged	✓	✓	✓	Exposed · on real LAN
Internal	✓	✗	✗	Isolated · no host
Host-only	✓	✓	✗	Isolated + host access

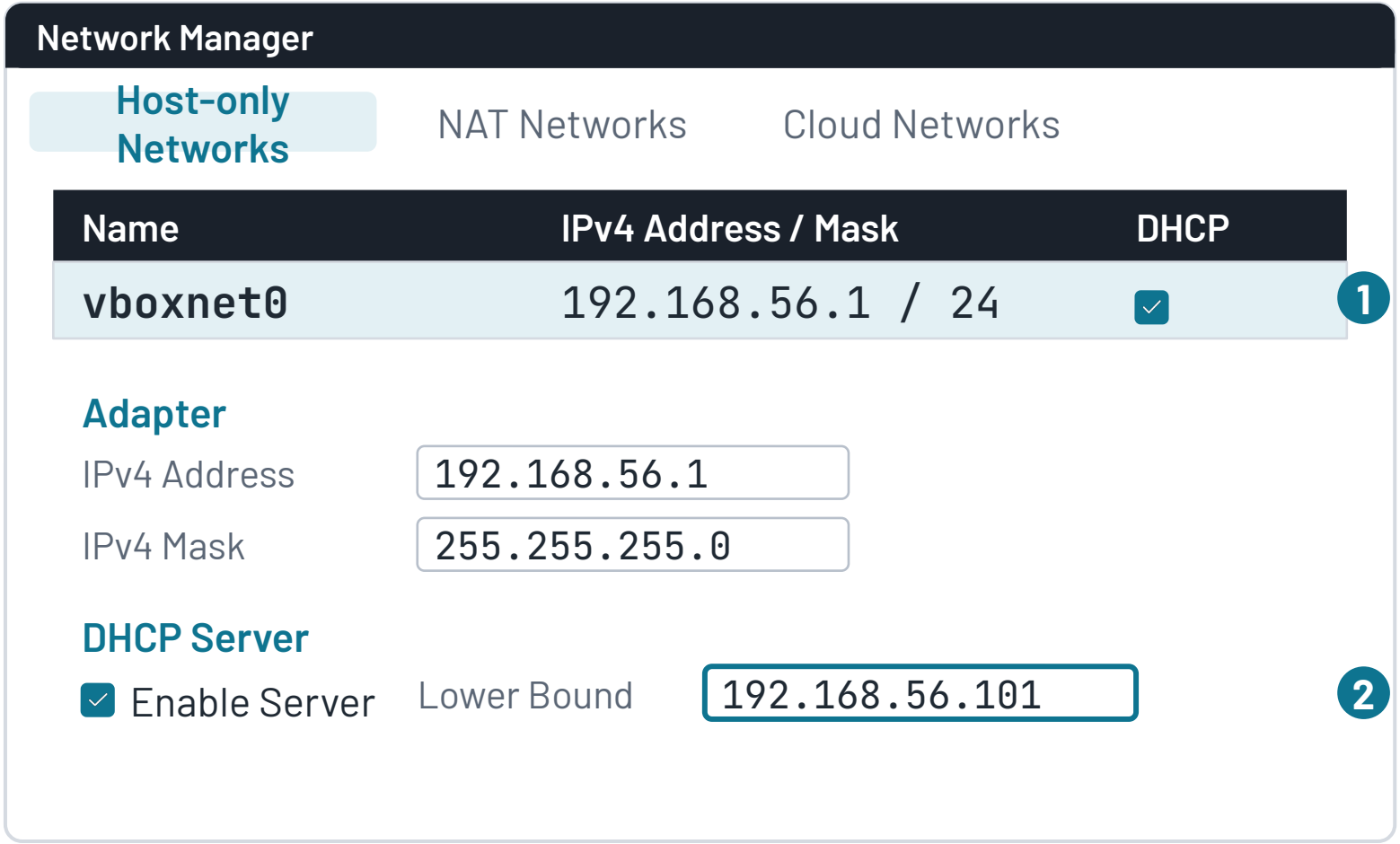
✓ Course choice: Host-only for the lab, plus NAT on a second Kali adapter for updates.

The adapter map you will build



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Create and check the Host-only network



1. **vboxnet0 exists**
   Default host-only network
2. **DHCP enabled**
   Hands out .101 upward
3. **Host is .1**
   Your gateway into the lab

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Point each adapter at the right network

Kali Linux · Network

Adapter 1 | Adapter 2 | Adapter 3 | Adapter 4

☑ Enable Network Adapter

Attached to: Host-only Adapter

Name: vboxnet0

```
Adapter 1 = NAT  (internet)
Adapter 2 = Host-only (lab)
```

Two adapters

Metasploitable 2 · Network

Adapter 1 | Adapter 2 | Adapter 3 | Adapter 4

☑ Enable Network Adapter

Attached to: Host-only Adapter

Name: vboxnet0

```
Adapter 1 = Host-only ONLY
Adapters 2-4 = disabled
```

One adapter, isolated

Prove the lab is wired correctly

kali@kali: ~

```
$ ip a show eth1
    inet 192.168.56.101/24 brd 192.168.56.255 scope global eth1
$ ping -c 4 192.168.56.102
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.42 ms
```

kali@kali: ~ · ping sweep

```
$ sudo nmap -sn 192.168.56.0/24
Nmap scan report for 192.168.56.1   (host)
Nmap scan report for 192.168.56.101 (kali)
Nmap scan report for 192.168.56.102 (target)
```


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When the lab won’t talk

SYMPTOM	LIKELY CAUSE	FIX
No IP address	Adapter off or DHCP disabled	**Enable adapter; turn on DHCP**
Can’t ping target	VMs on different networks	**Set both to the same vboxnet0**
No internet on Kali	NAT adapter disabled	**Enable Adapter 1 = NAT**

Quick check from Kali

```
$ ip a  ·  $ ping -c4 <target>  ·  $ ping -c4 kali.org
```

Snapshot both VMs as “Lab Ready”

Kali Linux



Clean · Lab Ready · Live

Updated · 2 adapters · verified

Metasploitable 2



Imported · Lab Ready · Live

Host-only · IP confirmed

Lab Ready
Revert here after every exercise to start clean and connected.

Isolate and verify the network



1. Confirm vboxnet0 and its DHCP server ☐
2. Kali: Adapter 1 NAT, Adapter 2 Host-only ☐
3. Metasploitable: Adapter 1 Host-only only ☐
4. From Kali, ping the target and sweep /24 ☐
5. Snapshot both VMs as “Lab Ready” ☐

>04 KEY TAKEAWAYS

What to remember from Lesson 04



+01

Host-only isolates the lab from your network



+02

NAT on a second Kali adapter brings updates

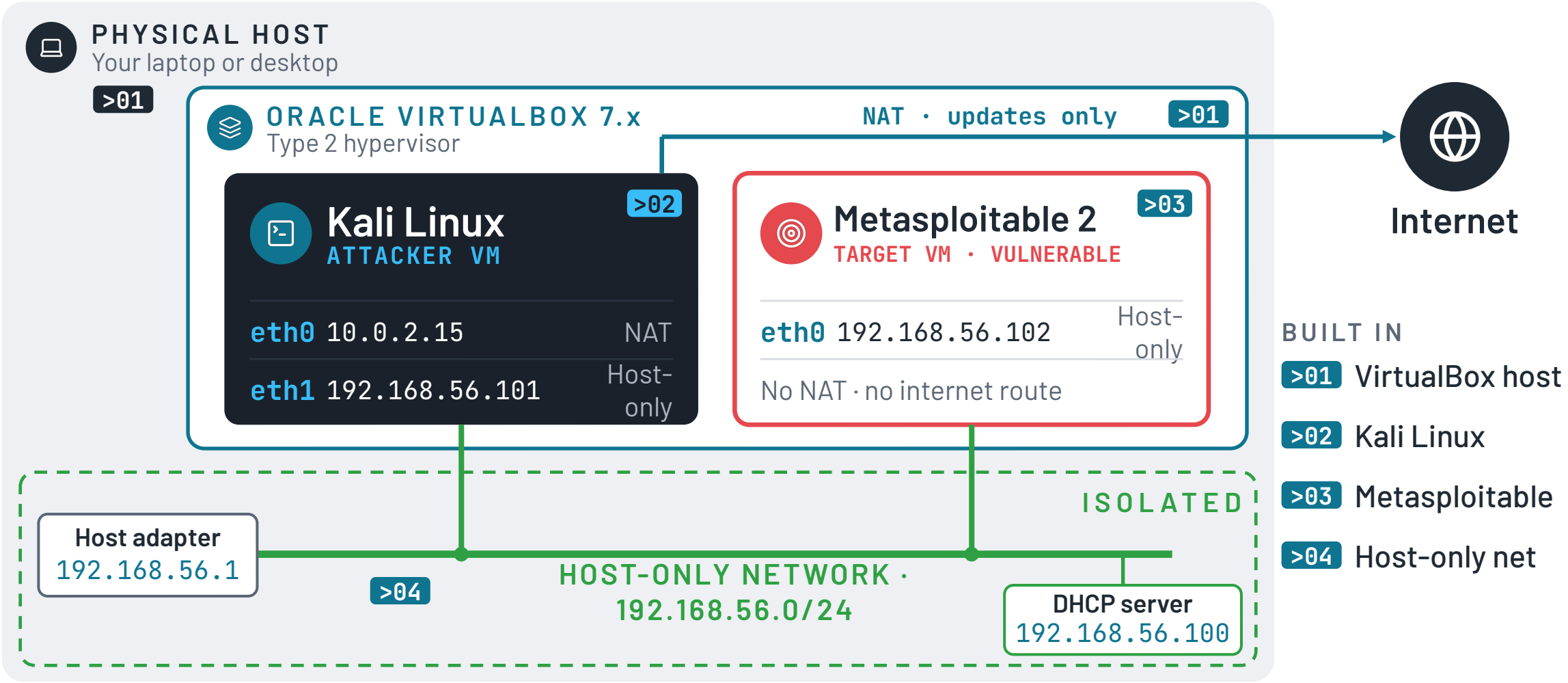


+03

Verify, then snapshot both VMs as Lab Ready

>_ MODULE SUMMARY

What you built, lesson by lesson

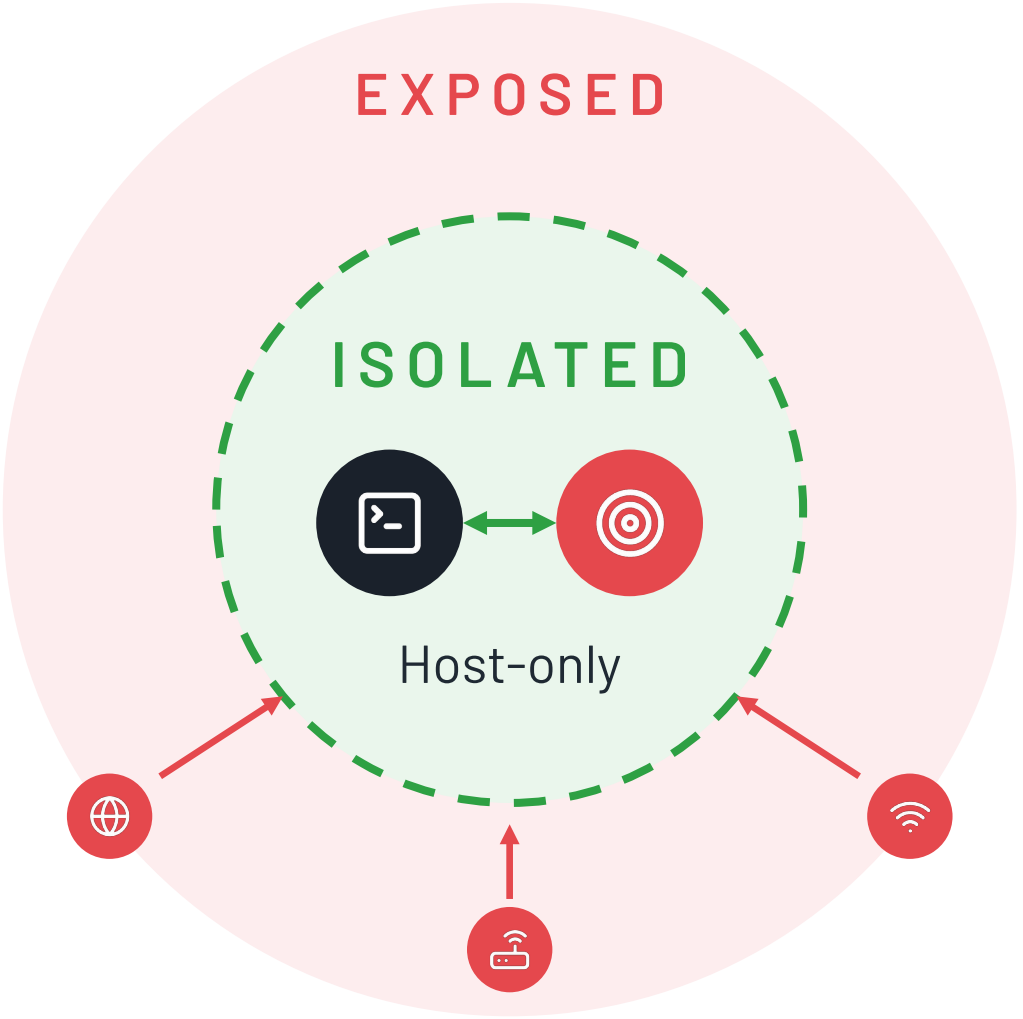


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>_ LAB SAFETY RULES

Four rules that keep your lab legal and contained

- Vulnerable VMs on Host-only
- Never bridge a target VM
- Attack only systems you own
- Revert targets after exercises



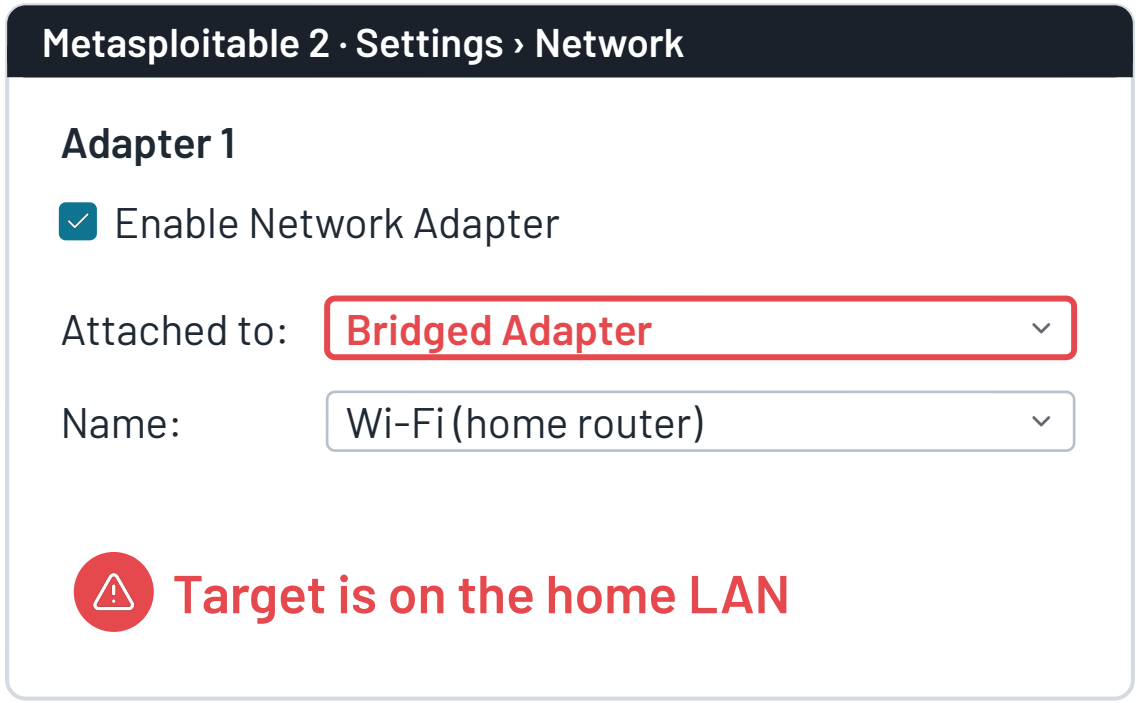
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>_ KNOWLEDGE CHECK · 1 / 5

Is this lab still safe?

Metasploitable's Adapter 1 was switched to Bridged so a teammate can scan it from their laptop.

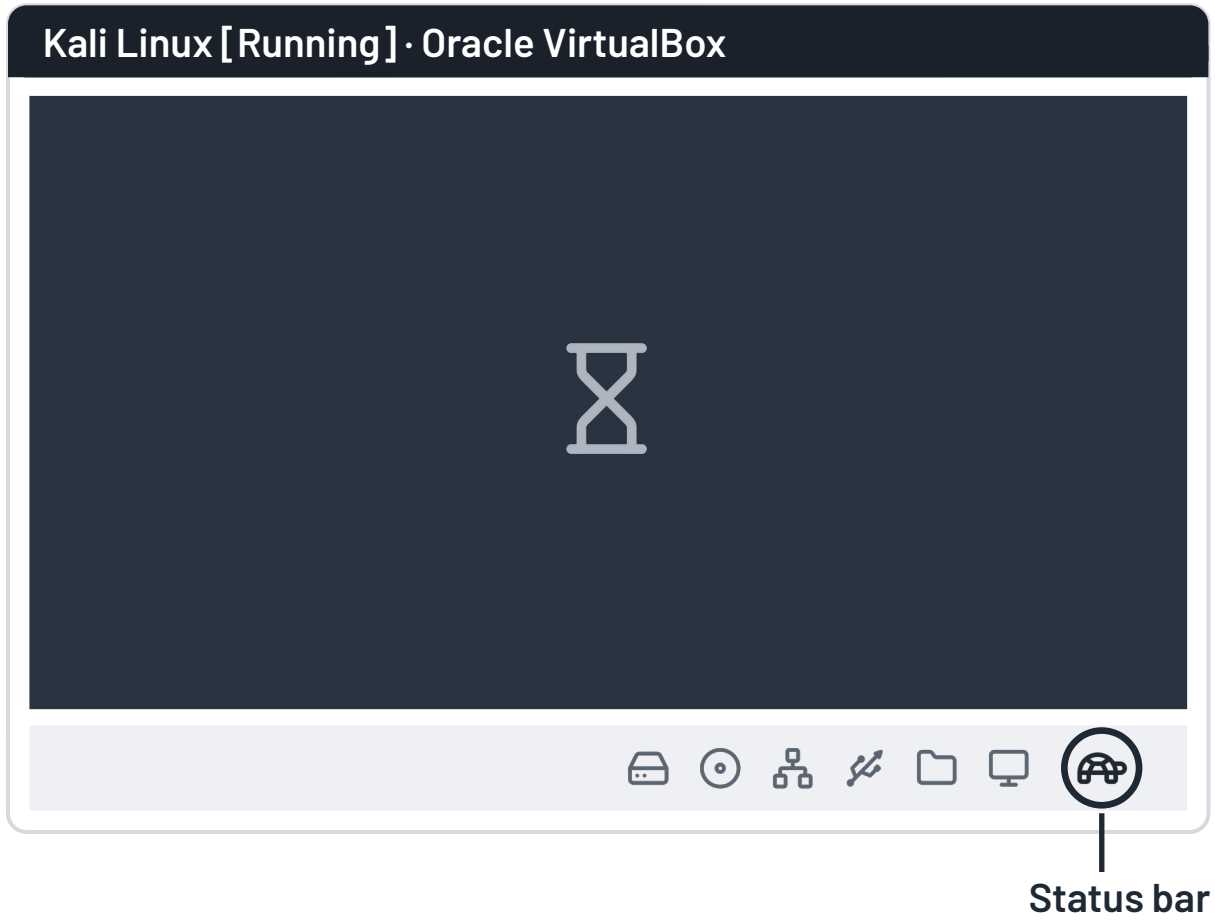
What is the risk, and what do you change?



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>_ KNOWLEDGE CHECK · 2 / 5

Why is every VM so slow?



Windows 11 host. VMs crawl, and a turtle icon appears in the VM status bar.

What is the cause, and how do you fix it?

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>_ KNOWLEDGE CHECK · 3 / 5

Where did Kali’s internet go?

Kali pings Metasploitable fine, but package updates fail.
Which adapter do you check first, and why?

kali@kali: ~

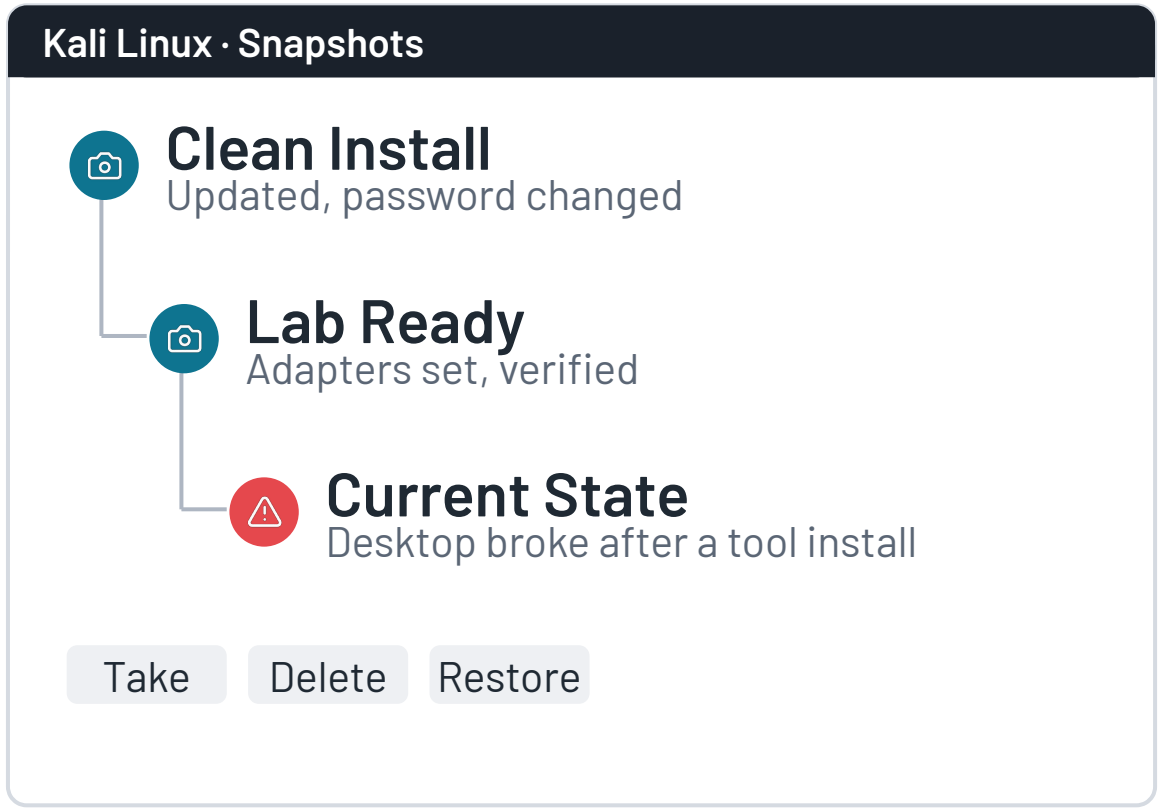
```
$ ping -c 1 192.168.56.102
1 packets transmitted, 1 received, 0% packet loss, time 0ms
```

kali@kali: ~

```
$ sudo apt update
Err:1 http://http.kali.org/kali kali-rolling InRelease
  Temporary failure resolving 'http.kali.org'
```

>_ KNOWLEDGE CHECK · 4 / 5

What is the fastest way back?



A tool install broke Kali's desktop. You snapshotted it after the first update.

How do you recover in under a minute?

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>_ KNOWLEDGE CHECK · 5 / 5

Where is the target?

Kali’s ping sweep finds the host adapter, but no Metasploitable.



Name two things you check first.

>_ PEHT · MODULE 02 COMPLETE

Next: Module 03

INSTRUCTOR

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